Concave and Convex Functions

ECON 441: Introduction to Mathematical Economics
Lecture 16

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Example

Find the critical points and determine whether each one is a maximum, a minimum, or neither.

$$f(x) = x^4 - 4x^3$$

Example

Find the critical points and determine whether each one is a maximum, a minimum, or neither.

$$f(x) = x + \frac{1}{x} \qquad (x \neq 0)$$

Concave and Convex Functions

- **Concave** function: $f''(x) \leq 0$ for all x
- **Convex** function: $f''(x) \geq 0$ for all x
- **Strictly concave** function: $f''(x) < 0$ for all x
- **Strictly convex** function: $f''(x) > 0$ for all x

Concave and Convex Functions

f is **concave** if, for any x_1, x_2 and $\lambda \in (0, 1)$,

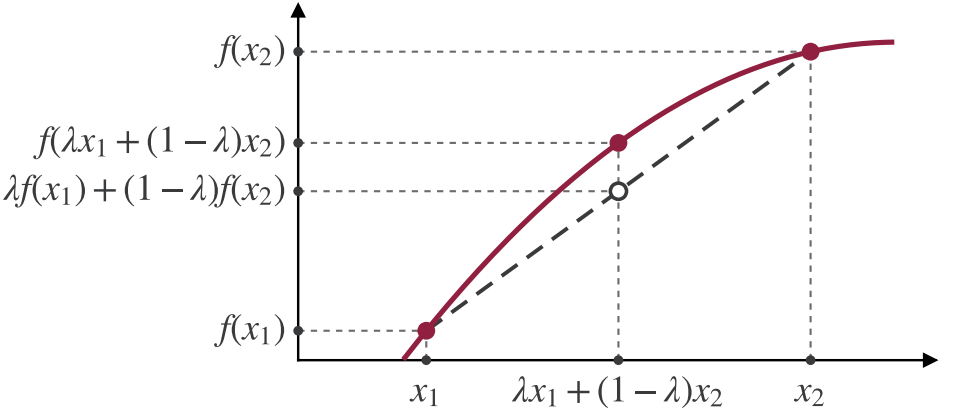
$$f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

f is **convex** if

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

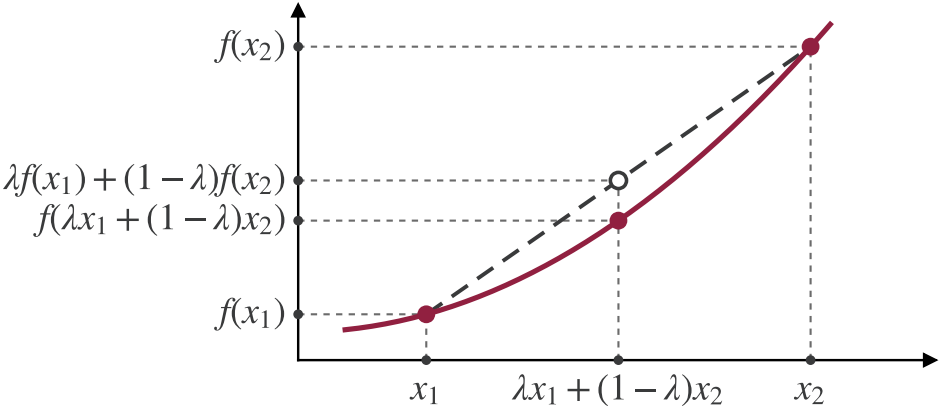
For strict concavity or convexity, replace with strict inequalities.

Concave Function



The curve lies **above** the chord.

Convex Function



The curve lies **below** the chord.

Attitudes toward Risk

Consider the following game: flip a coin, collect $0 if tails, collect $20 if heads. How much would you pay to play this game?

Example

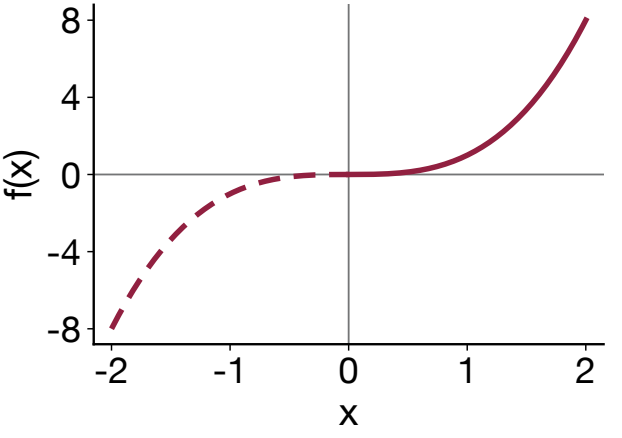
The domain of f is all real numbers:

$$f(x) = x^3$$

Is f convex, strictly convex, concave, or strictly concave?

What if the domain of f is all nonnegative real numbers?

$f(x) = x^3$



Properties of Concave and Convex Functions

1. A linear function is both concave and convex, but not strictly so.
2. If $f(x)$ is (strictly) concave, then $-f(x)$ is (strictly) convex, and vice versa.
3. If $f(x)$ and $g(x)$ are both concave (convex), then $f(x) + g(x)$ is concave (convex). If, in addition, one or both of them are strictly concave (convex), then $f(x) + g(x)$ is strictly concave (convex).

Global Optimizers

- If a function is concave, any critical point gives a global maximum.
- If a function is strictly concave, any critical point gives the **unique** global maximum.
- If a function is convex, any critical point gives a global minimum.
- If a function is strictly convex, any critical point gives the **unique** global minimum.

Example

Say $f(x)$ is a strictly concave function and

$$f'(2) = 0$$

Is $f(2)$ a local or a global maximum? Is it the unique maximum?

Example: Profit Maximization

A firm's profit is $\pi(Q) = R(Q) - C(Q)$ with revenue $R(Q) = 20Q - Q^2$ and cost $C(Q) = 4Q + 10$.

Find the profit-maximizing output. Is it a global maximum? Why?

Homework Questions

- Exercise 9.3: 2, 3, 4, 5
- Exercise 9.4: 5

Textbook reference: 9.3